Field components analysis inside the prototype magic angle $\theta_m \approx 54.74°$ helical solenoid

Axial component:

$$B_z = B_{\text{total}} \cos(54.74^\circ) \tag{1}$$

With

$$B_z = \mu_0 \mu_r n I \tag{2}$$

Where *n* is the number of turns per meter, μ_0 and μ_r the permeability of free space and relative permeability of the medium respectively and *I* is the electric current inside the coil (wire) of the solenoid.

Azimuthal component:

$$B_\phi = B_{\text{total}} \sin(54.74^\circ) \tag{3}$$

With

$$\tan(\theta_m) = \frac{B_\phi}{B_z} = \sqrt{2} \approx 1.41 \Rightarrow \theta_m = \arctan(\sqrt{2}) \approx 54.74^\circ. \tag{4}$$

Total field:

$$B_{\text{total}} = \sqrt{B_z^2 + B_\phi^2} \tag{5}$$

(This is the applied field when the solenoid is powered up without taking into account any residual magnetization of the ferrite core).

In our case,

$$\frac{B_\phi}{B_z} = \sqrt{2} \tag{6}$$

With

$$B_\phi = \sqrt{2} \cdot B_z \quad \text{and} \quad B_z = \frac{B_\phi}{\sqrt{2}} \tag{7}$$

Substituting to equation (5) we get,

$$B_z = \frac{B_{\text{total}}}{\sqrt{3}} \quad \text{and} \quad B_\phi = \sqrt{\frac{2}{3}} B_{\text{total}} \tag{8}$$